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Dihydrostreptamine Aminoglycosides: Neomycin Group

Neomycin is the isomeric mixture of neomycin B and C. Framycetin is identical to neomycin B. Paramomycin (aminosidine) is closely related to neomycin. Its pharmacokinetic and toxic properties and clinical applications are similar to those of neomycin so that, like neomycin, it is best reserved for local applications.

Antimicrobial Activity

Neomycin has similar activity to kanamycin on a weight basis and is several times more active than streptomycin; it is less active than gentamicin, tobramycin, and amikacin (see Table 10.1). Activity against *Staphylococcus aureus* is good, but it is generally low against other Gram-positive bacteria (see Tables 10.1 and 10.2). Many opportunist Gram-negative pathogens are susceptible to neomycin, although the prevalence of susceptible strains is slightly less than for kanamycin and far less than gentamicin. Many *Pseudomonas aeruginosa* are susceptible (see Table 10.2). Bacteria with MIC ≤ 8 $\mu\text{g/ml}$ are regarded as susceptible.

Resistance

Plasmid-mediated resistance occurs through a variety of enzymes (see Table 10.3). Such resistance, which is often multiple, is relatively common in enteric commensals and pathogens but less common among other opportunist pathogens. Chromosomal mutation to resistance is unimportant.

Drug Interactions

Neomycin shows synergistic activity with beta-lactam antibiotics and bacitracin against Gram-positive bacteria. Neomycin has sometimes been combined with macrolide antibiotics for intramammary treatment of

Gram-positive udder infections, but there is no information on the in vitro effect of the interaction.

Toxicity and Adverse Effects

Neomycin is the most toxic of the aminoglycosides and readily causes nephrotoxicity and deafness. For this reason it should never be used parenterally. Toxic effects are generally not produced when neomycin is administered orally or applied locally, but severe side effects of deafness and tubular necrosis have occurred in humans after oral administration.

Cats given high IM doses (100 mg/kg) daily showed nephrotoxic effects and became deaf in a few days; dogs are about equally susceptible. Total deafness was described in a dog after administration of 500 mg subcutaneously for 5 days (Fowler, 1968). In cattle, parenterally administered neomycin causes nephrotoxicity and deafness, which may be enhanced by dehydration. In pigs, transient posterior paresis and apnea immediately after injection have resulted from neuromuscular blockade. Malabsorption has been described in humans given oral neomycin for prolonged periods.

Administration and Dosage

Neomycin is reserved for local treatment of infections, often combined with bacitracin for broad-spectrum synergistic activity. In some countries neomycin is used as a parenteral injection for food animals and horses. Because of its toxic potential, such usage is strongly discouraged. Neomycin impaired the response of equine neutrophils to chemoattractants in vitro and in vivo, suggesting that the drug should be avoided for horses (Pycock et al., 1988). Paramomycin may be an effective treatment for acute cryptosporidiosis caused by *Cryptosporidium parvum* (Rehg, 1994; Mancassola et al., 1995) but may be less effective than azithromycin for this purpose. Whether neomycin might be equally efficacious remains to be determined. It should be noted that some neomycin appears to be absorbed following oral administration (approximately 3%) and may lead to kidney residues in cattle (Persoli et al., 1994).

Clinical Applications

Neomycin is used for the local treatment of intestinal infections, of wound or skin infections, and of mastitis. Its relatively broad spectrum of activity and the bactericidal effect made the drug popular in some countries for parenteral use in farm animals as an inexpensive "alternative" to gentamicin; safer, considerably less toxic, and more efficacious alternate drugs are now readily available.

CATTLE, SHEEP, AND GOATS

Neomycin has been used successfully in the oral treatment of enteric infections in calves, though resistance increasingly limits its effectiveness. One or 2 g administered orally twice daily has successfully controlled salmonellosis and enterotoxigenic *Escherichia coli* (ETEC) diarrhea in calves. Paramomycin and by implication neomycin might be efficacious in the treatment of cryptosporidiosis, but more work is required to investigate this point. On the other hand, Shull and Frederick (1978) found that routine addition of neomycin to milk powder of neonatal calves increased the frequency of diarrhea, possibly through a suppressive effect on normal intestinal microflora or through an irritant effect on the mucosa. Neomycin is absorbed after oral administration and can lead to violative kidney drug residues. Routine intrauterine administration of neomycin boluses to postparturient cattle was followed by significantly increased number of services per conception compared to controls (Fuquay et al., 1975).

In dry-cow treatment, neomycin alone has given poor cure rates, but combination with penicillin G has produced excellent cure of staphylococcal and streptococcal infections, far superior to penicillin-streptomycin combinations (LeLouedec, 1978). Neomycin has been combined with spiramycin for use in dry-cow treatment with reasonable effect. There is evidence that neomycin dry-cow preparations, particularly those also containing corticosteroids, have predisposed cows to *Nocardia* mastitis. It is unclear if this is a general property of neomycin-corticosteroids or was the result of contamination of batches (Ollis et al., 1991).

Paromomycin is available in some countries for parenteral use in food animals, with one application being the treatment of bovine respiratory disease (Brugere-Picoux et al., 1986).

SWINE

Neomycin is used in the oral treatment of *E. coli* diarrhea, although resistance increasingly limits its use.

HORSES

Neomycin may be used in the local treatment of infections caused by susceptible bacteria. It has been used in Australia in the parenteral treatment of *Rhodococcus equi* pneumonia in foals, often administered for prolonged periods without evidence of toxicity (Barton, 1986), but better alternatives are readily available. Evidence of nephrotoxicity was observed in adult horses within 4 days of parenteral administration (Edwards et al., 1989).

DOGS AND CATS

Neomycin may be used for the local treatment of infections in dogs and cats, such as otitis externa and anal sac infections.

POULTRY

Neomycin is sometimes administered orally to chickens and turkeys in the treatment of *Salmonella* infections.

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Kanamycin Group

The kanamycin group contains the kanamycins and semisynthetic derivatives such as amikacin; the nebramycins such as tobramycin and apramycin; and gentamicin, netilmicin, and sisomicin.

Kanamycin

ANTIMICROBIAL ACTIVITY

Kanamycin (Fig. 10.2) has similar activity to neomycin. It is active against many species of mycobacteria and mycoplasma but is inactive